GROUP 13

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We will predict if the students will **Pass the Exam (Yes/No)** based on these five specific factors.

Student	Review	Study Hours	Sleep	Arrival	Pass the exam?
1	Yes	High	Good	Early	Yes
2	No	Low	Poor	Late	No
3	Yes	Low	Poor	Early	Yes
4	No	High	Good	Early	No
5	Yes	High	Good	Early	Yes
6	No	Low	Good	Late	No
7	Yes	High	Poor	Early	Yes
8	No	High	Good	Early	No
9	Yes	High	Poor	Early	Yes
10	Yes	Low	Good	Early	Yes

Outlook:

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Review	Y	N	P(Yes)	P(No)
Yes	6	0	6/6	0/4
No	0	4	0/6	4/4
Total	6	4		

Study hours	Y	N	P(Yes)	P(No)
High	4	2	4/6	2/4
Low	2	2	2/6	2/4
Total	6	4		

Sleep	Y	N	P(Yes)	P(No)
Good	3	3	3/6	3/4
Poor	3	1	3/6	1/4
Total	6	4		

Arrival	Y	N	P(Yes)	P(No)
Early	6	2	6/6	2/4
Late	0	2	0/6	2/4
Total	6	4		

Solving:

$$P(y|x_1...x_n) = P(y)\prod_{i=1}^{n} P(x_i|y)$$

(Review=Yes, Study hours=High, Sleep = Good, Arrival= Early)

P(Pass the exam? = YES) = 6/10 = .6

P(y) = $\frac{6}{10}(\frac{6}{6}x\frac{4}{6}x\frac{3}{6}x\frac{6}{6})$
P(y) = (0.6)(0.33)
P(y) = 0.198

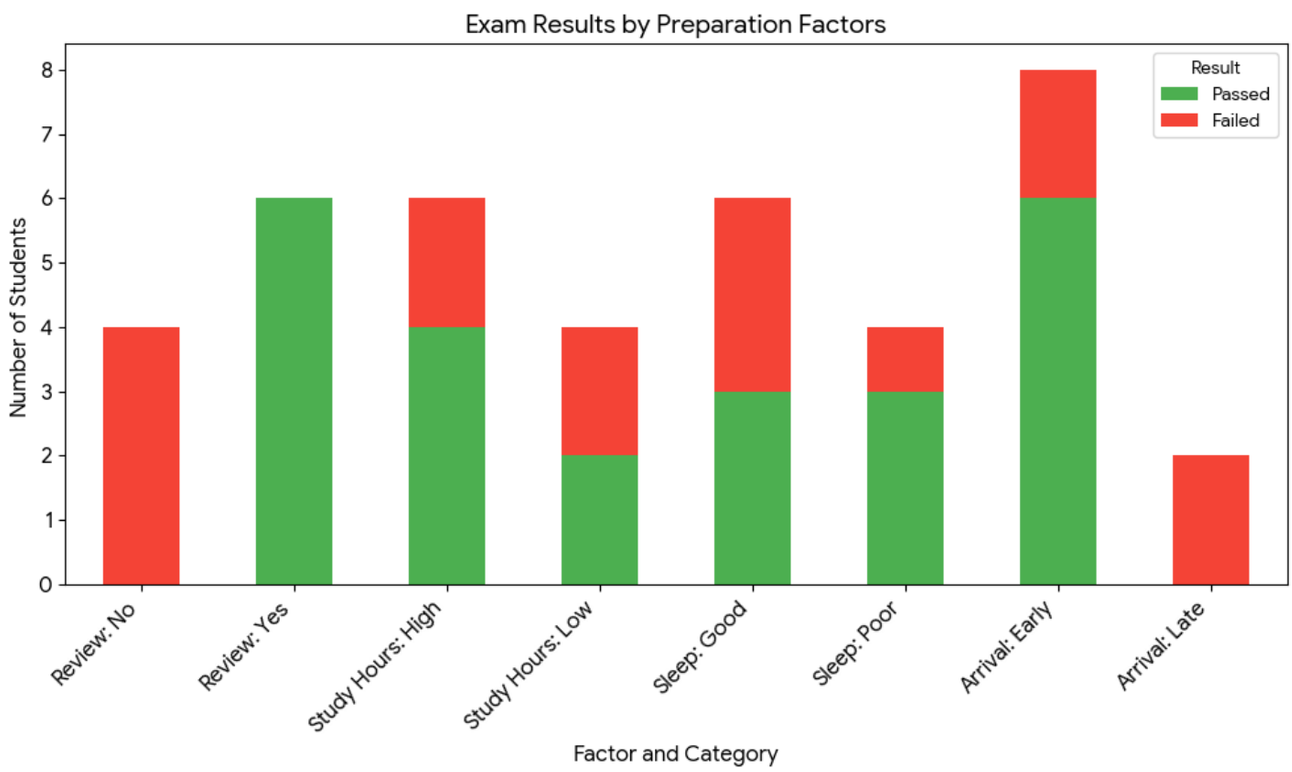
P(Pass the exam? = NO) = 4/10 = .4

P(n) = $\frac{4}{10}(\frac{0}{4}x\frac{2}{4}x\frac{3}{4}x\frac{2}{4})$
P(n) = (0.4)(0)
P(n) = 0

Conclusion:
In our dataset, the Review and Arrival factors are 'perfect predictors.' Because there are zero instances of a student failing after attending the review, the Naive Bayes formula multiplies the 'Fail' probability by zero, resulting in a prediction of a guaranteed Pass.

- Pass Score: 0.198
- Fail Score: 0.0

Prediction: PASS. Since 0.198 is greater than 0, the model predicts the students will pass the exam.

Comparison of Influence

Factors	Predicts Passing?	Why?
Review	Extremely High	No one in this data passed without it.
Study Hours	Very High	All passing students were on time.
Sleep	Moderate	Some students with "Low" hours still passed.
Arrival	Low	Passing students were split 50/50 between Good and Poor sleep.